FATIMA JINNAH WOMEN UNIVERSITY, RAWALPINDI

DEPARTMENT OF COMPUTER SCIENCE

LAB 02

Working with Git History, Stashing, and Reverting Commits

Submitted by

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DevOps (CS-649)

BCS VII

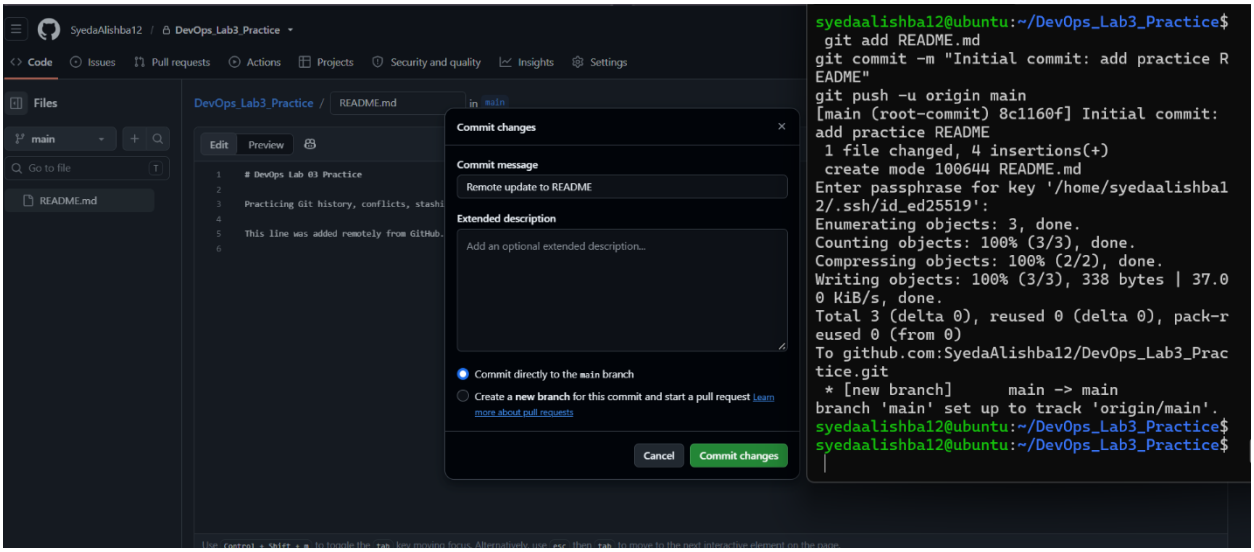
Github Username

SyedaAlishba12

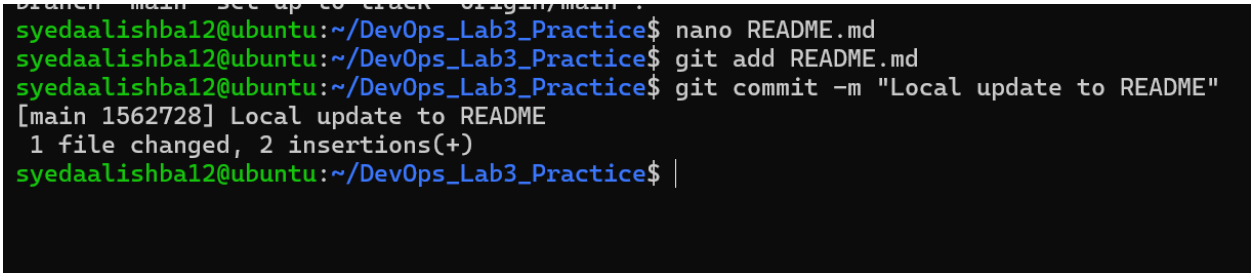
Lab 03: Working with Git History, Stashing, and Reverting Commits

Task 1: Handling Local and Remote Commit Conflicts

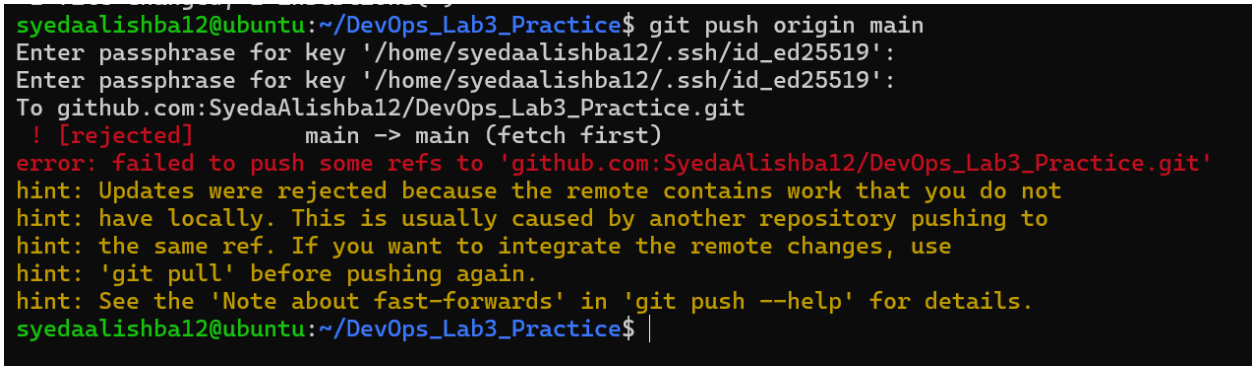
This task demonstrated how local and remote changes can cause push conflicts. I practiced resolving them using merge and rebase.



Change made on GitHub.



Local changes committed.



Push rejected due to remote changes.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 627 bytes | 313.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   0765bfb..7df2359  main -> main
branch 'main' set up to track 'origin/main'.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

Changes pushed after merging.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 951 bytes | 158.00 KiB/s, done.
From github.com:SyedaAlishba12/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   7df2359..171a74a  main       -> origin/main
Successfully rebased and updated refs/heads/main.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

Remote changes integrated using rebase.

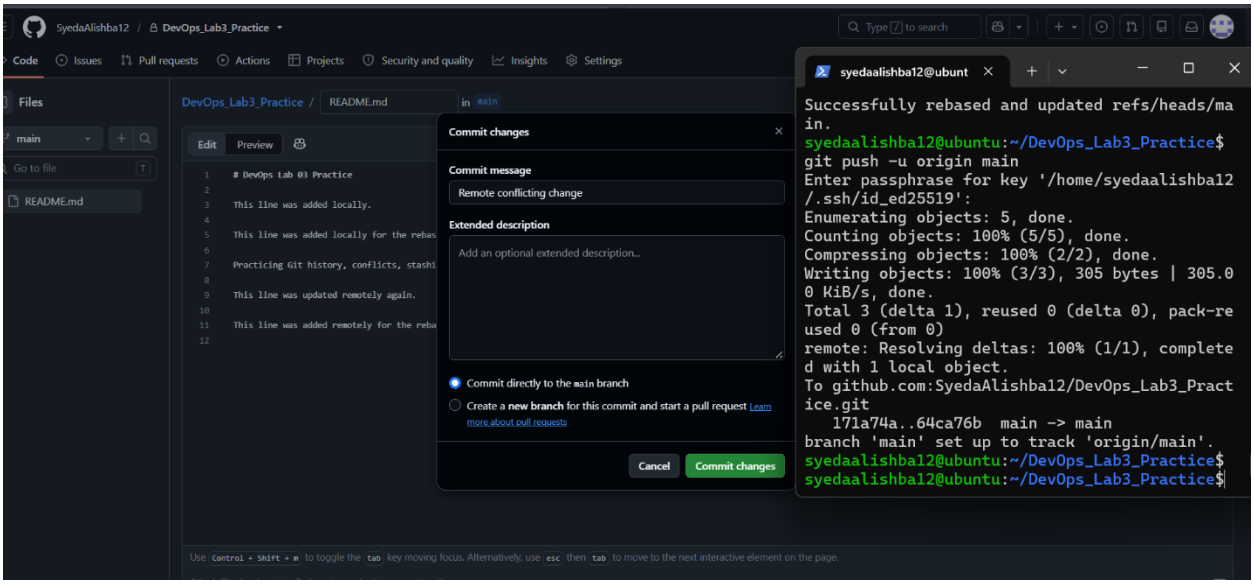
```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 305 bytes | 305.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   171a74a..64ca76b  main -> main
branch 'main' set up to track 'origin/main'.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

Changes pushed after rebase.

Task 2: Creating and Resolving Merge Conflicts Manually

This task demonstrated how Git handles different changes to the same line. I resolved the conflict and pushed the updated file.

Screenshots:

- 

Conflicting change made remotely.

- ```
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local conflicting change"
  [main 09e46d0] Local conflicting change
   1 file changed, 1 insertion(+), 1 deletion(-)
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
  ```

Local conflicting change committed.

- ```
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
  Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
  To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   ! [rejected]        main -> main (fetch first)
  error: failed to push some refs to 'github.com:SyedaAlishba12/DevOps_Lab3_Practice.git'
  hint: Updates were rejected because the remote contains work that you do not
  hint: have locally. This is usually caused by another repository pushing to
  hint: the same ref. If you want to integrate the remote changes, use
  hint: 'git pull' before pushing again.
  hint: See the 'Note about fast-forwards' in 'git push --help' for details.
  syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
  ```

Push rejected because of conflicting history.

```
 GNU nano 8.7.1                                README.md
# DevOps Lab 03 Practice

This line was added locally.

This line was added locally for the rebase exercise.

Practicing Git history, conflicts, stashing, resetting, and remote collaboration.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for the rebase exercise.

                          [ Read 12 lines ]
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```

- Conflict resolved in the README.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 328 bytes | 328.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   da044d7..9a38d60  main -> main
branch 'main' set up to track 'origin/main'.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Resolved changes pushed successfully.

Task 3: Managing Ignored Files with .gitignore

This task demonstrated how .gitignore manages files Git should ignore and how previously tracked files can be removed from tracking.

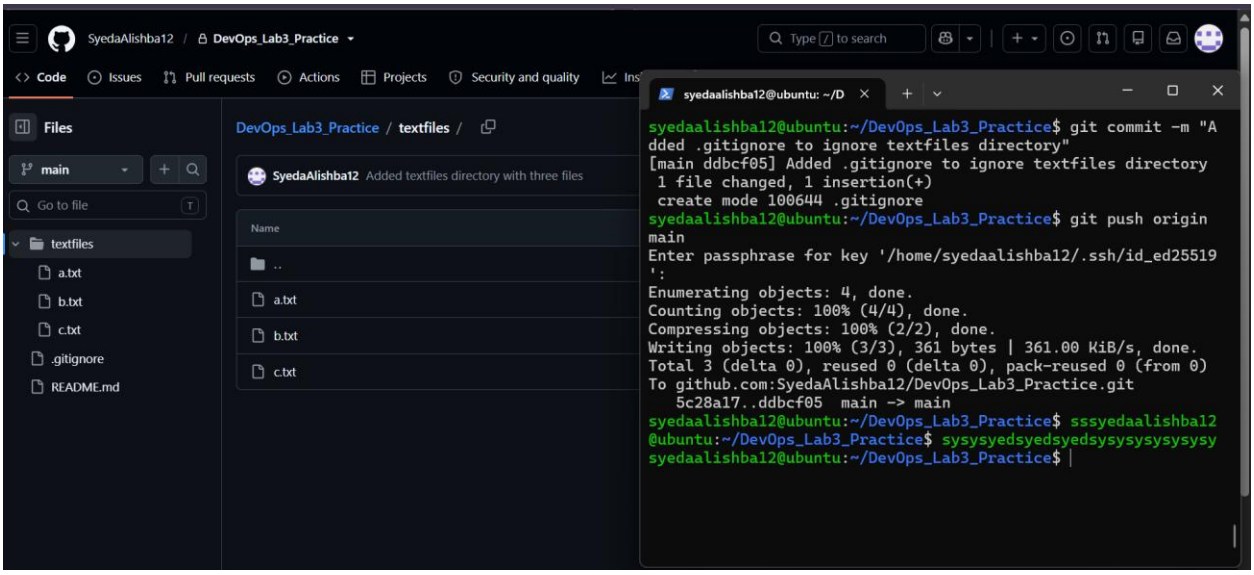
Screenshots:

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ echo "textfiles/*" > .gitignore
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main ddbcf05] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 361 bytes | 361.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   5c28a17..ddbcf05  main -> main
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

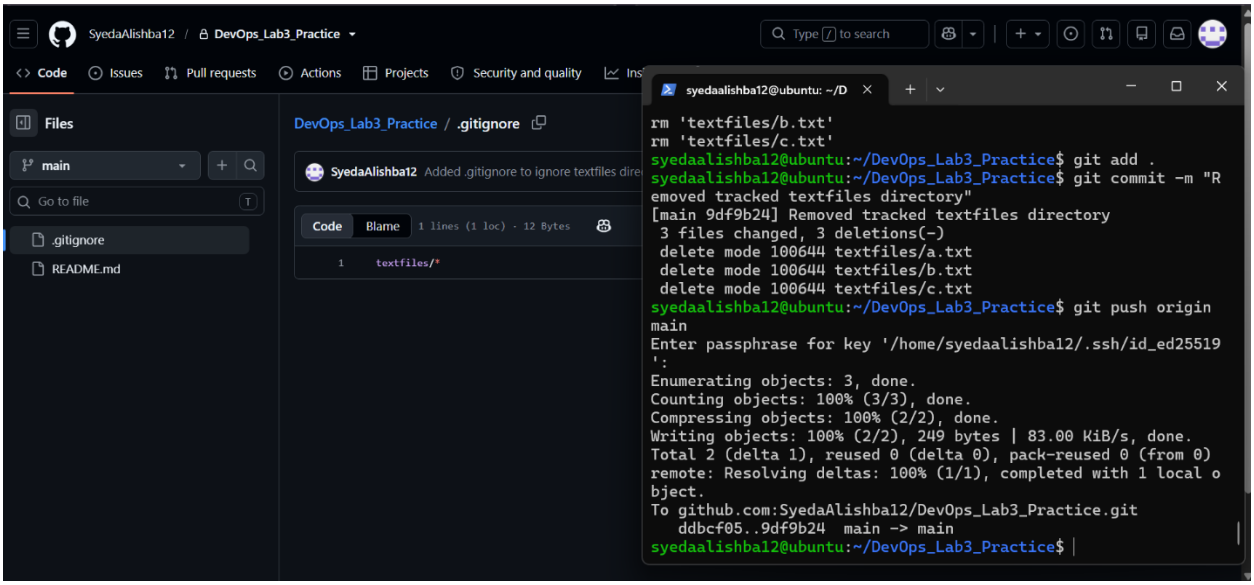
- .gitignore changes pushed.

```
hba12@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git add .
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Removed tracked textfiles directory"
[main 9df9b24] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 249 bytes | 83.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
   ddbcf05..9df9b24  main -> main
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Previously tracked files removed from tracking.



- Text files still present locally or in the repository before removal.

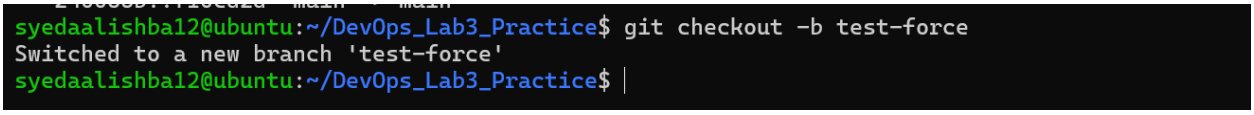


- Repository after the tracked text files were removed.

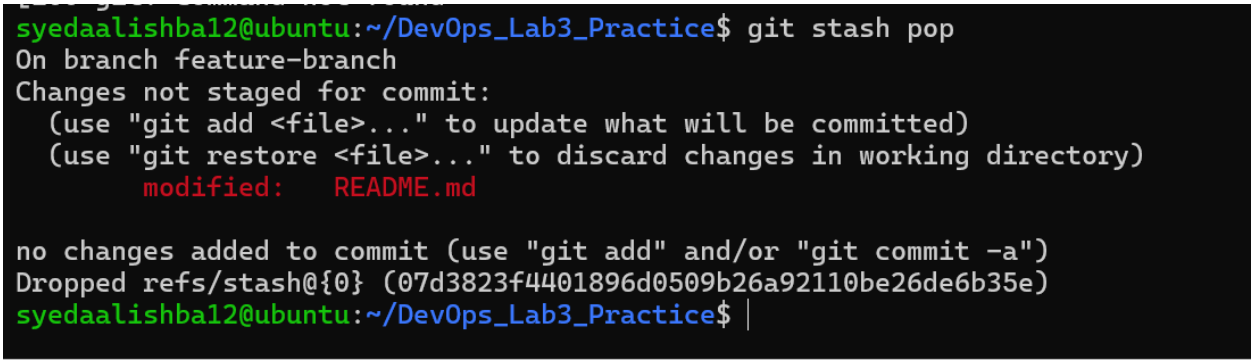
Task 4: Creating Temporary Changes and Using Git Stash

This task demonstrated how to save unfinished changes temporarily and restore them when needed.

Screenshots:



- New branch created for the exercise.



- Stashed changes restored.

Task 5: Checking Out a Specific Commit

This task demonstrated how to inspect previous commits and return to the required branch.

Screenshots:

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
867bfdd (HEAD -> feature-branch) Changes the README.md file
9df9b24 (origin/main, main) Removed tracked textfiles directory
ddbcf05 Added .gitignore to ignore textfiles directory
5c28a17 Added textfiles directory with three files
9a38d60 Local conflicting change
da044d7 Remote conflicting change
64ca76b Local update for rebase
171a74a Remote update for rebase
7df2359 Merge branch 'main' of github.com:SyedaAlishba12/DevOps_Lab3_Practice
1562728 Local update to README
0765bfb Remote update to README
04ad77b Remote update to README
8c1160f Initial commit: add practice README
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Commit history before checkout.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git checkout 9df9b24
Note: switching to '9df9b24'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 9df9b24 Removed tracked textfiles directory
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- A specific commit checked out.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Returned to the feature branch.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Returned to the main branch.

Task 6: Resetting Commits — Soft and Hard Reset

This task demonstrated the difference between soft and hard resets and their effects on the working files and commit history.

Screenshots:

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 03 Practice

This line was added locally.

This line was added locally for the rebase exercise.

Practicing Git history, conflicts, stashing, resetting, and remote collaboration.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for the rebase exercise.

First test line for reset exercise.

Second test line for reset exercise.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- File before resetting.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Soft reset performed.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 03 Practice

This line was added locally.

This line was added locally for the rebase exercise.

Practicing Git history, conflicts, stashing, resetting, and remote collaboration.

This line was updated remotely again.
This line was updated locally at the same time.

This line was added remotely for the rebase exercise.

First test line for reset exercise.

Second test line for reset exercise.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- File after soft reset.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
497f405 (HEAD -> main) Second test commit
a9db428 Added test line
9df9b24 (origin/main) Removed tracked textfiles directory
ddbcf05 Added .gitignore to ignore textfiles directory
5c28a17 Added textfiles directory with three files
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
497f405 (HEAD -> main) Second test commit
a9db428 Added test line
9df9b24 (origin/main) Removed tracked textfiles directory
ddbcf05 Added .gitignore to ignore textfiles directory
5c28a17 Added textfiles directory with three files
9a38d60 Local conflicting change
da044d7 Remote conflicting change
64ca76b Local update for rebase
171a74a Remote update for rebase
7df2359 Merge branch 'main' of github.com:SyedaAlishba12/DevOps_Lab3_Practice
1562728 Local update to README
0765bfb Remote update to README
04ad77b Remote update to README
8c1160f Initial commit: add practice README
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Commit history before reset.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at a9db428 Added test line
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Hard reset performed.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second Test commit"
[main 3cbd0c4] Second Test commit
 1 file changed, 2 insertions(+)
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- File after hard reset.

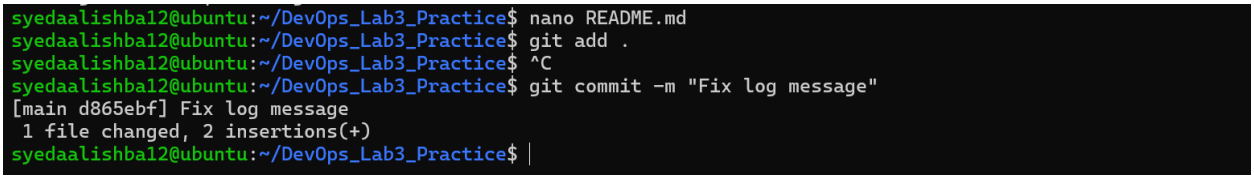
```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
a9db428 (HEAD -> main) Added test line
9df9b24 (origin/main) Removed tracked textfiles directory
ddbcf05 Added .gitignore to ignore textfiles directory
5c28a17 Added textfiles directory with three files
9a38d60 Local conflicting change
da044d7 Remote conflicting change
64ca76b Local update for rebase
171a74a Remote update for rebase
7df2359 Merge branch 'main' of github.com:SyedaAlishba12/DevOps_Lab3_Practice
1562728 Local update to README
0765bfb Remote update to README
04ad77b Remote update to README
8c1160f Initial commit: add practice README
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

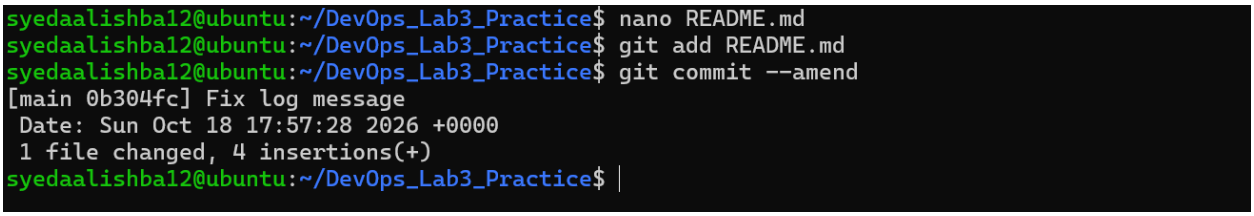
- Commit history after hard reset.

Task 7: Amending the Last Commit

This task demonstrated how to update the most recent commit instead of creating a separate commit.

Screenshots:

- 

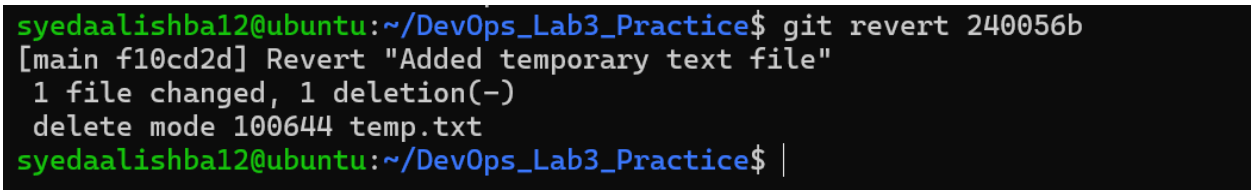
  Initial commit before amendment.
- 

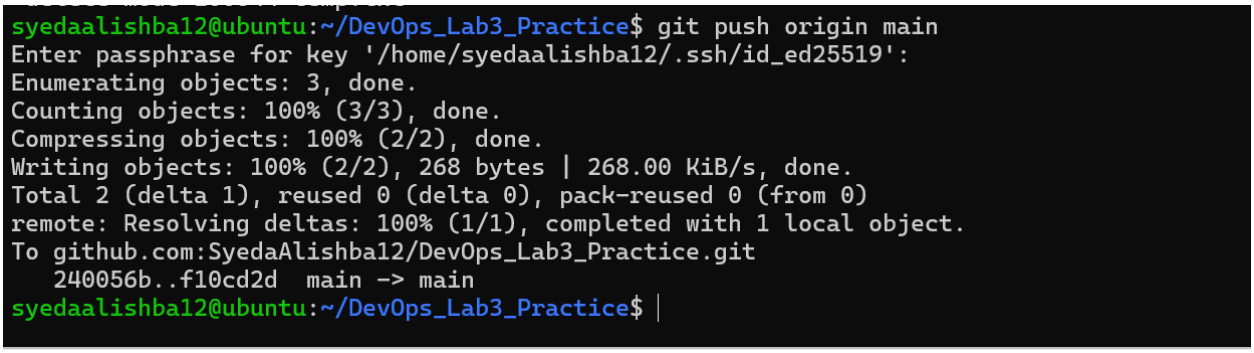
  Updated commit after amendment.

Task 8: Reverting a Commit

This task demonstrated how to undo a previous commit safely by creating a new revert commit.

Screenshots:

- 

  Revert commit created.
- 

  Revert commit pushed to the remote repository.

Task 9: Force Push with Caution

This task demonstrated how force pushing updates remote history after a local reset. The exercise was performed on a separate test branch.

Screenshots:

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ echo "Force push practice line" >> README.md
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Add force push practice line"
[test-force d1e5954] Add force push practice line
 1 file changed, 1 insertion(+)
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Commit created on the test branch.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin test-force
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 354 bytes | 177.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
branch 'test-force' set up to track 'origin/test-force'.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Test branch pushed to GitHub.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at f10cd2d Revert "Added temporary text file"
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Branch reset to an earlier commit.

```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:SyedaAlishba12/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Normal push rejected.

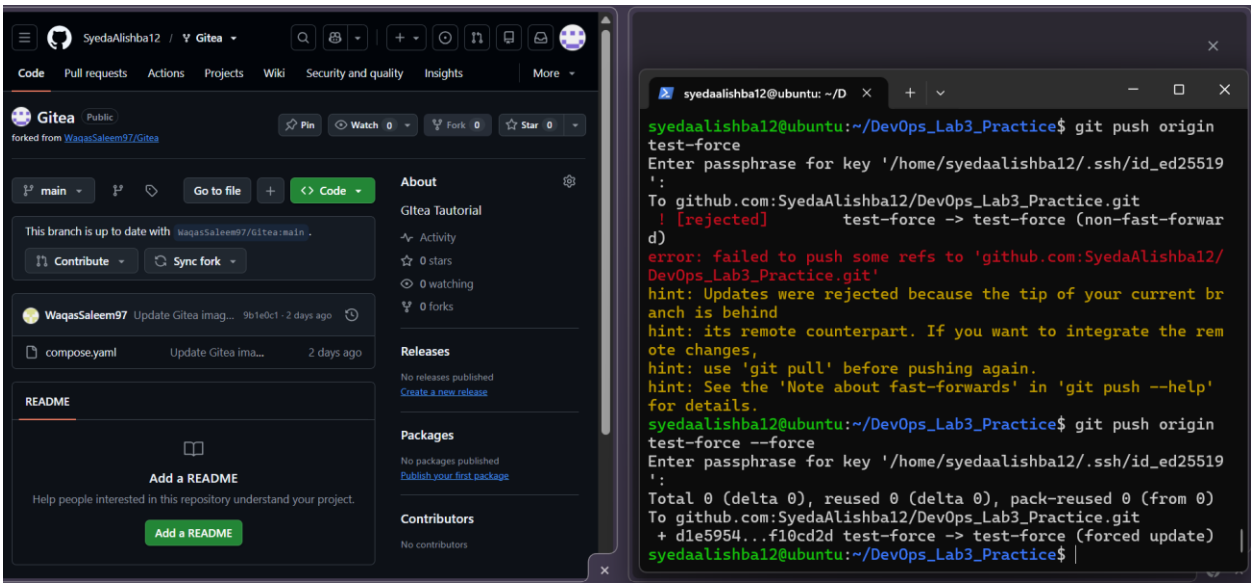
```
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:SyedaAlishba12/DevOps_Lab3_Practice.git
 + d1e5954...f10cd2d test-force -> test-force (forced update)
syedaalishba12@ubuntu:~/DevOps_Lab3_Practice$
```

- Force push completed.

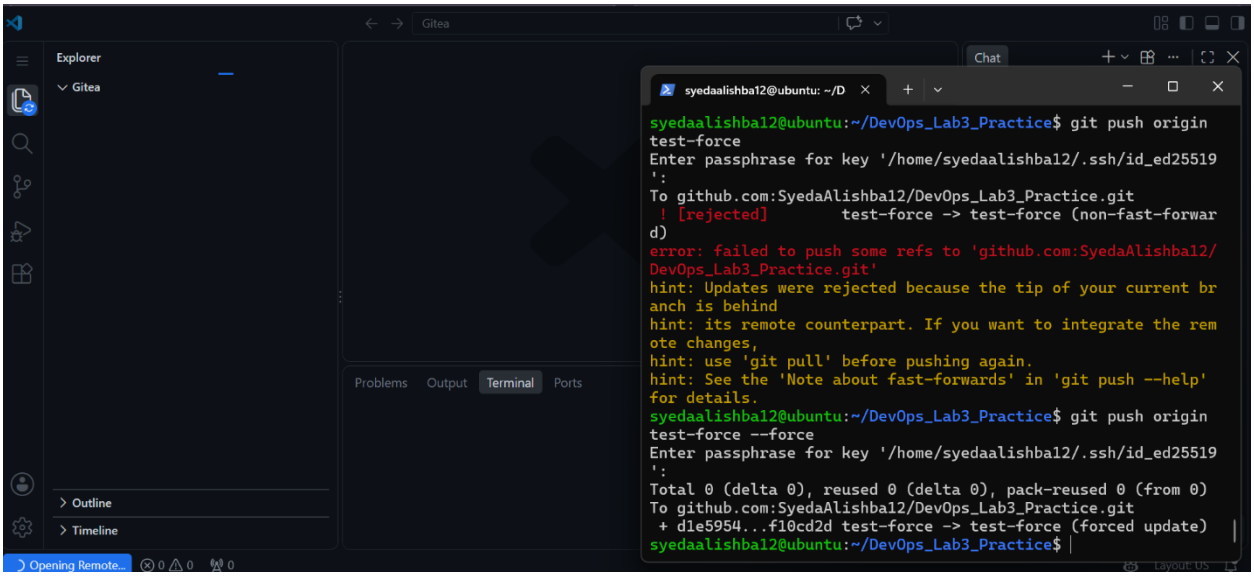
Task 10: Running Gitea in GitHub Codespaces

This task demonstrated how to run Gitea and PostgreSQL using Docker Compose in GitHub Codespaces.

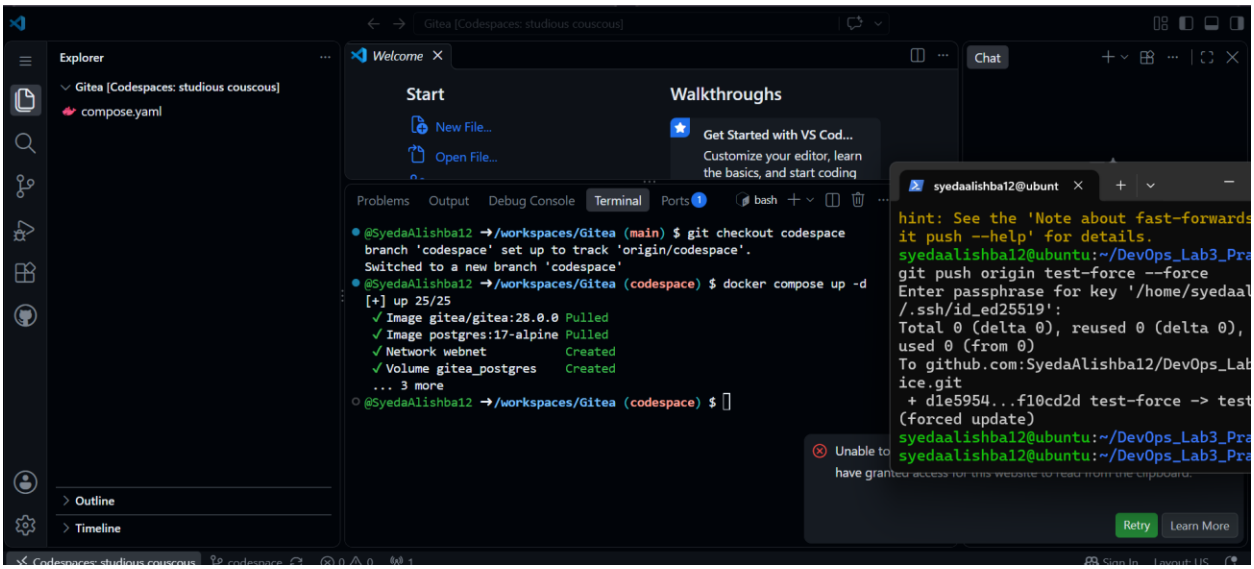
Screenshots:

- 

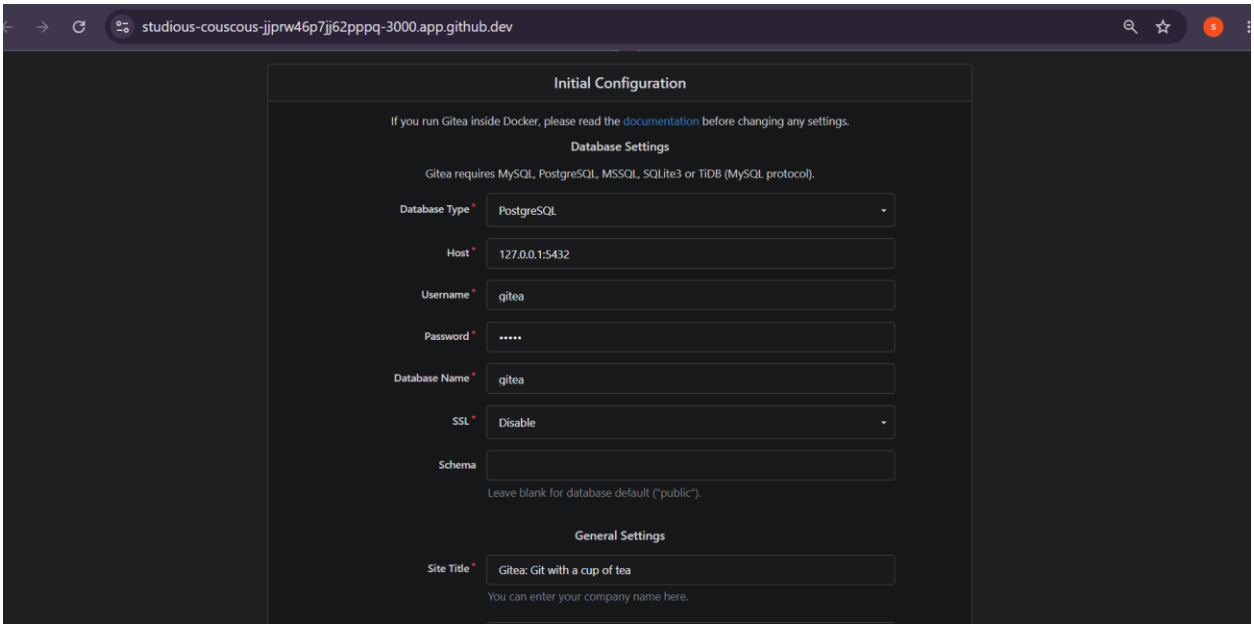
Gitea repository forked.

- 

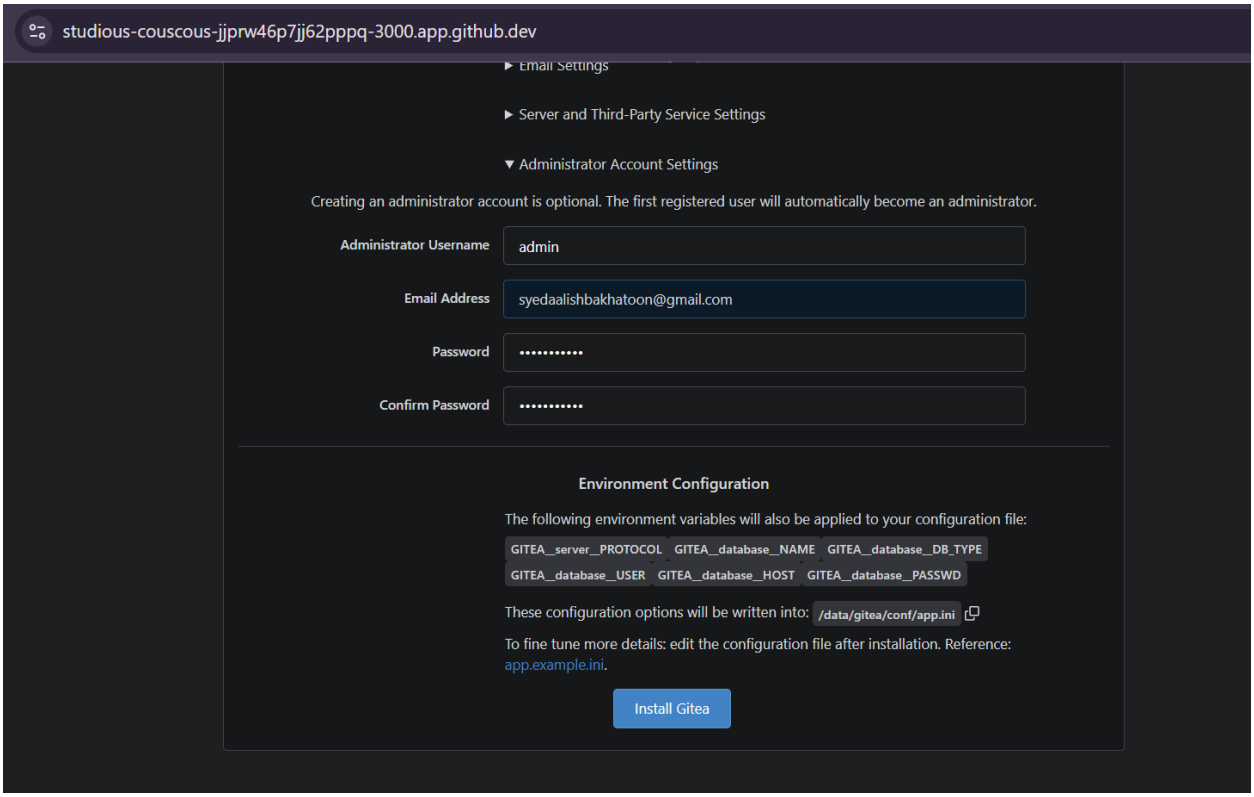
Codespace opened.

- 

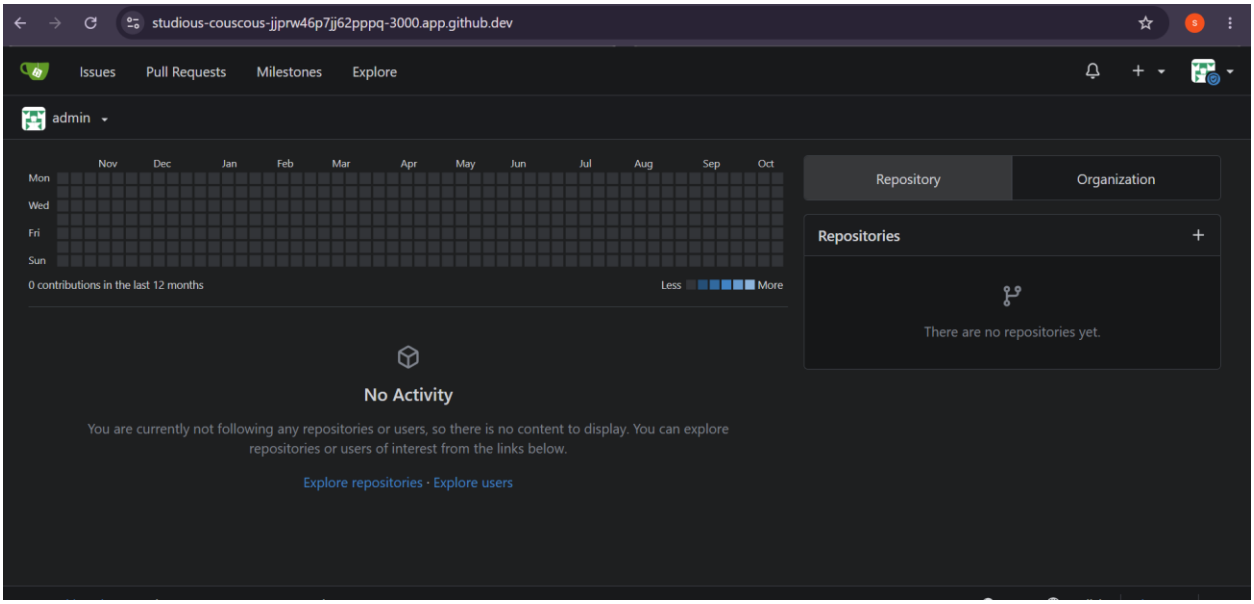
Docker Compose services started.

- 

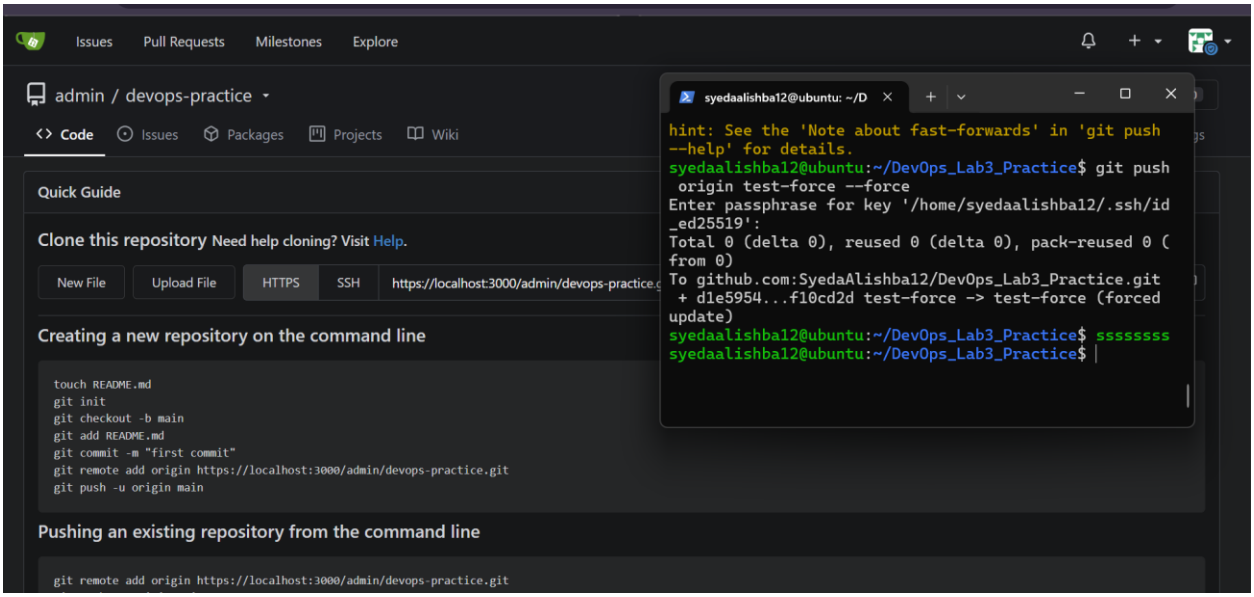
Gitea installation page.



- Initial administrator setup.



- Gitea dashboard.

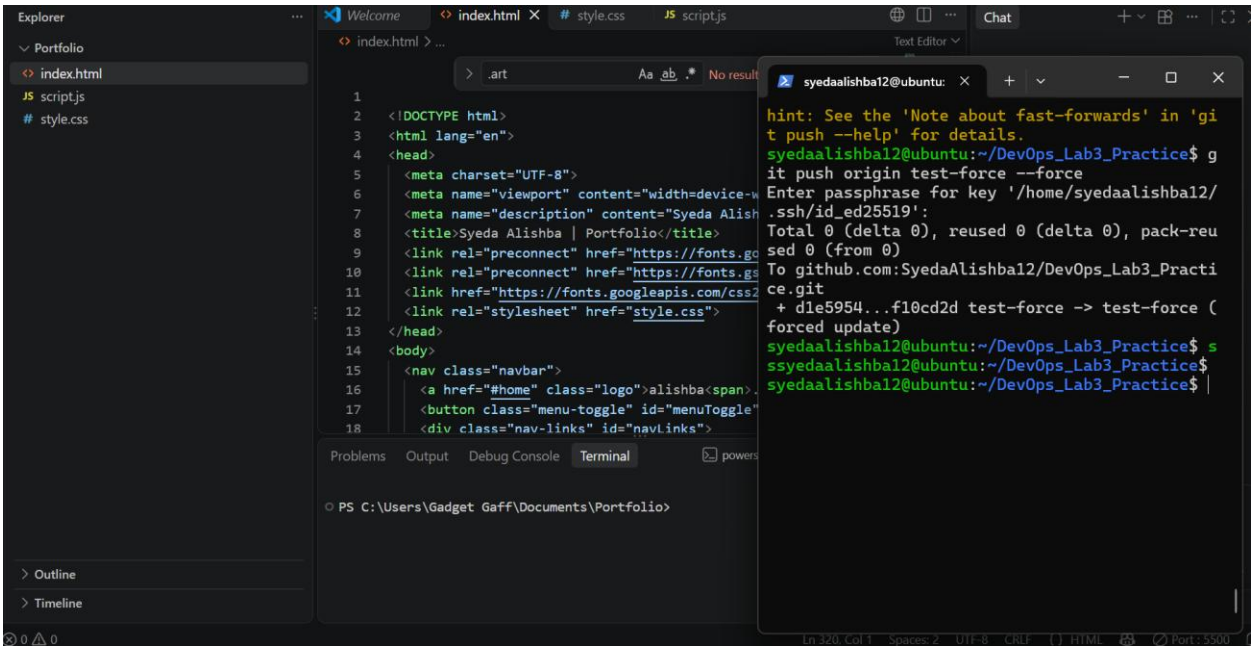


- Repository created in Gitea.

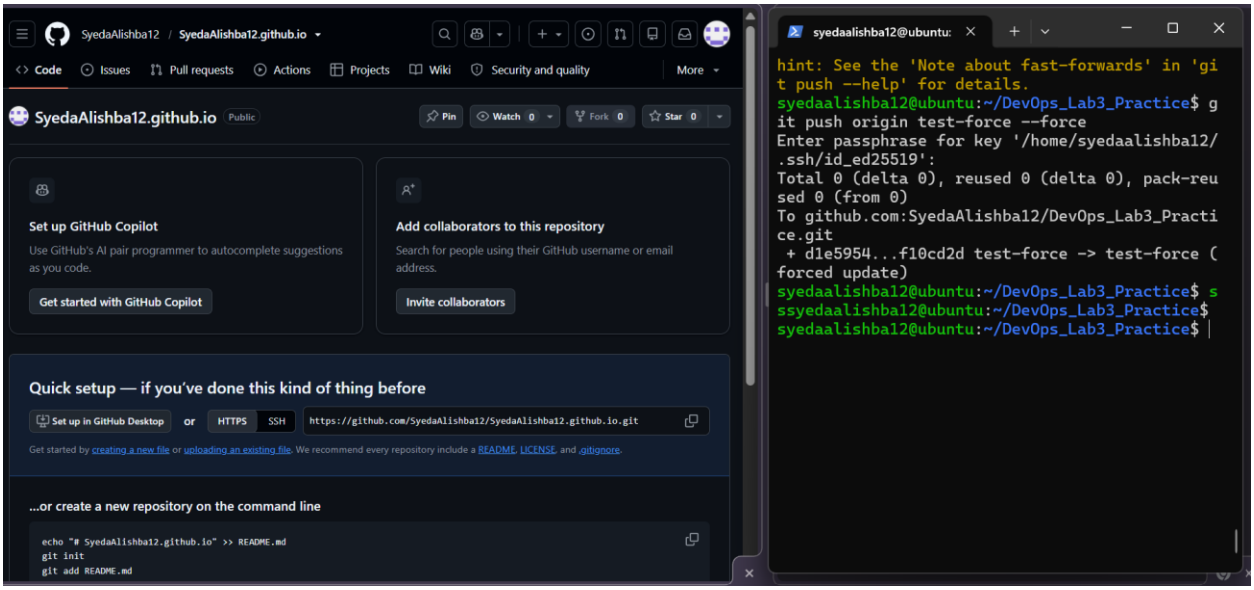
Task 11: Creating a GitHub Pages Portfolio Site

This task demonstrated how to publish a static portfolio website through GitHub Pages.

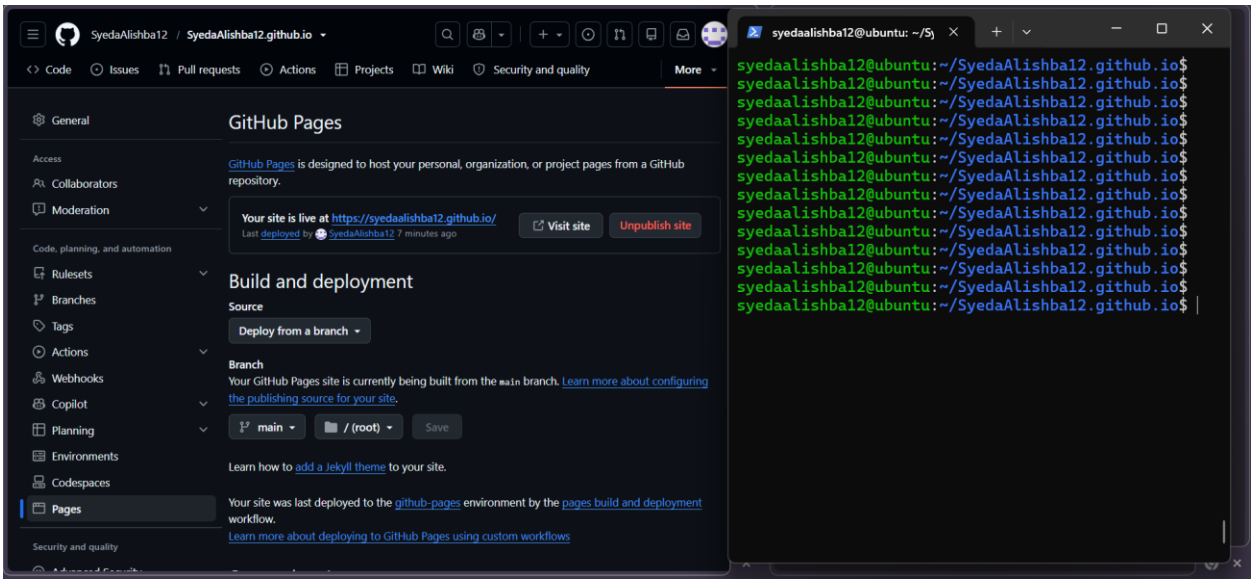
Screenshots:



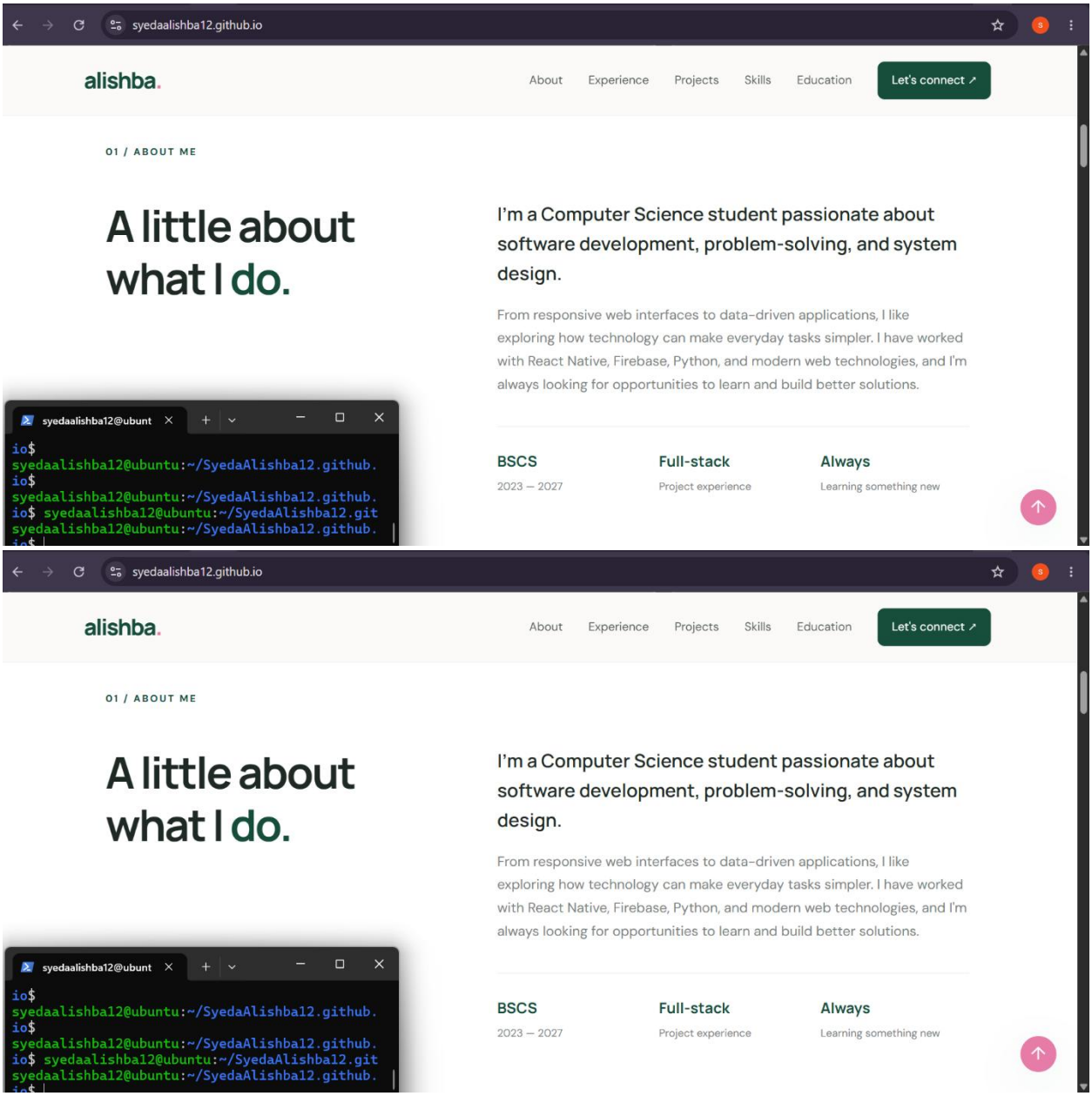
- Portfolio website files on the local computer.

- 

Portfolio repository on GitHub.

- 

GitHub Pages configuration.



Published portfolio website.

Portfolio URL: https://SyedaAlishba12.github.io

Conclusion

In this lab, I practiced Git conflict resolution, stashing, commit history management, resets, amendments, reverting, and force pushing. I also ran Gitea with Docker Compose and published a portfolio website using GitHub Pages.